Spectral Clustering

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1 Introduction

Clustering is a fundamental problem in data analysis and machine learning, whose objective is to partition a set of data points into groups - called clusters - such that elements within the same group are more similar to each other than to those belonging to different groups.
Graph clustering is the task of grouping the vertices of the graph into clusters taking into consideration the edge structure of the graph. In this framework, data points are represented as nodes and their similarities as edges, allowing the use of tools from graph theory and linear algebra. Furthermore, graph representation captures global relationships in the data that standard distance-based methods cannot see.

2 k-means Clustering

The k-means algorithm is widely used in data science because it is both simple to understand and computationally efficient. Given a fixed number of clusters k, the main goal of the algorithm is to partition the data by minimizing the within-cluster variance. The process is iterative: it starts with an initial set of centroids and then alternates between two steps.
First, it assigns each data point to the nearest centroid, usually based on the Euclidean distance, and then it updates the centroids by calculating the mean of all points in each cluster.
This procedure repeats until convergence, which is the point where the cluster assignments and the centroids no longer change.
Because it is fast and easy to implement, k-means is especially useful for large datasets and is often chosen as a baseline method for clustering tasks. When applied to a dataset, the algorithm organizes all points into k groups by minimizing the total internal variance across the entire set.

2.1 Limitations of k-means

Despite its advantages, k-means suffers from several important limitations. The algorithm relies on Euclidean distance so assumes that clusters are convex and well separated.
As a result, it fails to correctly identify clusters with complex geometry, such as non-convex or non-linearly separable structures.
A second limitation concerns anisotropic datasets, where clusters exhibit elongated shapes or significantly different variances along different directions.
In these cases, k-means tends to produce partitions that do not reflect the true structure of the data, since the distance to the centroid is not an adequate measure of similarity. These limitations motivate the use of graph-based approaches, such as Spectral Clustering.

Figure 1: Failure of k-means on non-convex and anisotropic datasets

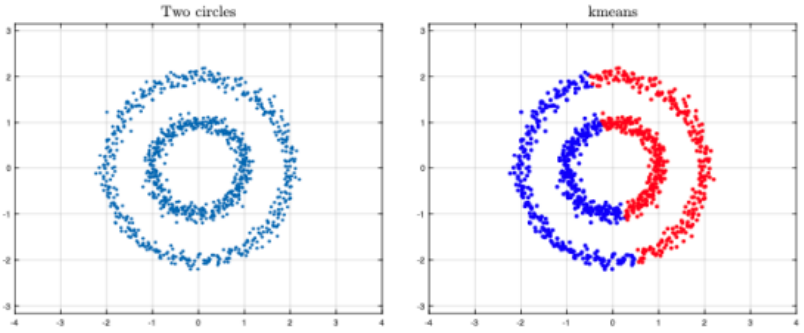


Figure: Data on two concentric circles

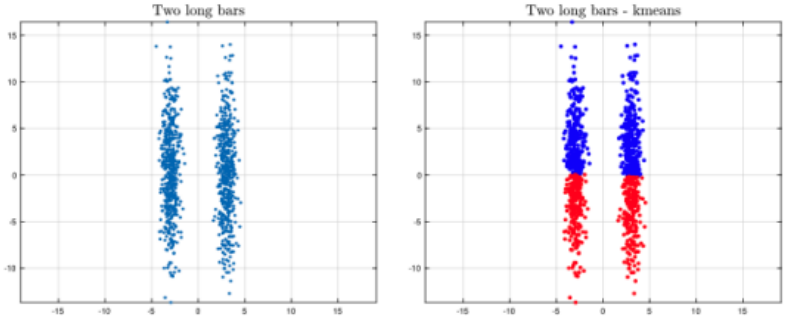


Figure: Anisotropic data

2.2 Spectral Clustering

Spectral clustering is a graph-based clustering method that exploits the spectral properties of matrices associated with a graph representation of the data. Unlike classical clustering techniques that only rely on distances in the original feature space, spectral clustering captures the global structure of the data by modeling pairwise similarities through a graph.
Each data point is represented as a vertex, while weighted edges encode the similarity between pairs of points. By analyzing the eigenvalues and eigenvectors of the graph Laplacian, the method produces a low-dimensional representation in which clusters become more easily separable.
This approach is particularly effective in identifying non-convex clusters and complex geometric structures, where distance-based methods such as k-means typically fail.

2.2.1 Laplacian graph

Let $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^d$ be a set of data points. Spectral clustering starts by building a graph $G = (V, E)$ where each vertex $v_i \in V$ is represented by a node x_i. In this model, the edge (i, j) between two nodes has the weight w_{ij} shows how similar those two points are to each other. A standard choice for the similarity function is the Gaussian (RBF) kernel:

$$w_{ij} = \begin{cases} \exp\left(-\dfrac{\|x_i - x_j\|^2}{2\sigma^2}\right), & i \neq j, \\ 0, & i = j, \end{cases}$$

where $\sigma > 0$ is a scale parameter controlling the locality of the similarities. Small values of σ emphasize local relationships, while larger values produce a more densely connected graph.
Other choices to define $w_{i,j}$ are ϵ-neighborhood, which connects all vertices v_i, v_j such that $d_{i,j} < \epsilon$ and the knearest neighbor graph (connect v_i to its k nearest neighbors).
The similarity weights define the adjacency matrix

$$W = (w_{ij}) \in \mathbb{R}^{n \times n}, \qquad w_{ij} \geq 0,$$

from which the diagonal degree matrix $D \in \mathbb{R}^{n \times n}$ is constructed as

$$D_{ii} = \sum_{j=1}^{n} w_{ij}.$$

Finally, the graph Laplacian L is derived from W and D, with different definitions depending on the chosen normalization. The unnormalized graph Laplacian is defined as

$$L = D - W.$$

3 Properties of L

Let $L = D - W$ denote the unnormalized graph Laplacian of an undirected weighted graph.

L is symmetric and positive semi-definite.

All eigenvalues of L are real and non-negative.

The smallest eigenvalue is $\lambda_1 = 0$, with associated eigenvector given by the constant vector $\mathbf{1}$.
An important property of the graph Laplacian concerns the eigenvalue $\lambda = 0$:

- The multiplicity of $\lambda = 0$ equals the number of connected components of the graph.
- A vector x satisfies $Lx = 0$ if and only if it is constant on each connected component.

These properties show that the eigenvectors corresponding to the smallest eigenvalues of L encode information about the connectivity structure of the graph.

3.1 Normalized Graph Laplacians

In addition to the unnormalized graph Laplacian, normalized variants are commonly used. Let D be the degree matrix and W the adjacency matrix.

- **Symmetric normalized Laplacian:**

$$L_{\text{sym}} = D^{-1/2} L D^{-1/2} = I - D^{-1/2} W D^{-1/2}.$$

- **Random-walk normalized Laplacian:**

$$L_{\text{rw}} = D^{-1} L = I - D^{-1} W.$$

Both L_{sym} and L_{rw} are positive semi-definite and have real, non-negative eigenvalues.

3.2 Properties of Normalized Laplacians

The normalized graph Laplacians L_{sym} and L_{rw} share the following properties:

- The smallest eigenvalue of both matrices is 0.
- The eigenvector associated with $\lambda = 0$ is $D^{1/2}\mathbf{1}$ for L_{sym} and $\mathbf{1}$ for L_{rw}.

The eigenvectors of L_{sym} and L_{rw} are directly related:

- If (λ, u) is an eigenpair of L_{rw}, then $D^{1/2}u$ is an eigenvector of L_{sym} associated with the same eigenvalue λ.

3.3 Choice of K

Determining the optimal number of clusters k is a central challenge in clustering. In spectral clustering, we can exploit the spectral properties of the Laplacian matrix to find a suitable k. The **Eigengap Heuristic** suggests to choose k such that all eigenvalues $\lambda_1, \ldots, \lambda_k$ are very small, while λ_{k+1} is relatively large. This is based on the perturbation theory: if a graph has exactly k connected components, then $\lambda_1 = \cdots = \lambda_k = 0$ and $\lambda_{k+1} > 0$. When the components are not perfectly disconnected but only weakly linked, the first k eigenvalues will be close to zero, and a significant "gap" will appear between λ_k and λ_{k+1}.

4 Algorithm of Spectral Clustering

Given a set of data points, spectral clustering is performed through the following steps.

1. Construct an undirected weighted graph whose vertices correspond to the data points and whose edge weights encode pairwise similarities.
2. Compute the degree matrix D, with entries $D_{ii} = \sum_j w_{ij}$, and form the unnormalized graph Laplacian
$$L = D - W.$$
3. Compute the eigenvectors $\phi_1, \ldots, \phi_k$ associated with the k smallest eigenvalues of L.
4. Construct the matrix $U \in \mathbb{R}^{n \times k}$ by stacking the selected eigenvectors as columns, and represent each data point by the corresponding row of U.
5. Cluster the rows of U into k groups using the k-means algorithm.

4.1 Normalized Spectral Clustering

The spectral clustering procedure can also be defined using normalized Laplacians.
In this case, Step 2 of the algorithm is modified by replacing L with L_{sym}.
The remaining steps (eigenvector computation, spectral embedding, and final clustering via k-means) follow the same structure.
The main difference between unnormalized and normalized spectral clustering lies in how node connectivity (degree) influences the spectral embedding.

- **Unnormalized spectral clustering** is based on the Laplacian $L = D - W$. In this case, nodes with large degree have a stronger influence on the eigenvectors. This approach is suitable when the graph is relatively regular and node degrees do not vary significantly.
- **Normalized spectral clustering** uses a normalized Laplacian, such as L_{sym}. The normalization rescales each node by its degree, reducing the dominance of highly connected nodes. As a result, the spectral embedding is less affected by differences in node connectivity and better reflects the underlying cluster structure.

The spectral clustering procedure does not directly produce a partition of the data. Instead, it provides a low-dimensional representation, known as the spectral embedding, in which the geometry of the data is significantly simplified. In this space, points belonging to the same cluster are mapped close to each other, while points from different clusters are well separated. As a result, a simple distance-based clustering method such as k-means becomes effective when applied to the rows of U, even in cases where k-means fails in the original data space.

5 Experimental Results

The dataset consists of three concentric circular clusters embedded in $\mathbb{R}^2$. The inner circle contains 100 points with radius approximately equal to 1, the middle circle contains 200 points with radius approximately equal to 3, and the outer circle contains 300 points with radius approximately equal to 5. A small Gaussian noise with standard deviation 0.05 is added to the radial component of each point. This configuration produces clusters that are clearly non-convex and characterized by different densities, making the dataset unsuitable for classical distance-based clustering methods such as k-means applied in the original space.

5.1 Graph Construction and Spectral Embedding

A fully connected weighted graph is constructed using a Gaussian similarity function

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right), \quad w_{ii} = 0,$$

with scale parameter $\sigma = 0.3$. From the affinity matrix W and the degree matrix D, two different graph Laplacians are considered: the unnormalized Laplacian and the symmetric normalized Laplacian.
In both cases, the first $k = 3$ eigenvectors associated with the smallest eigenvalues are computed and used to construct a low-dimensional spectral embedding. The final clustering is obtained by applying the k-means algorithm to the rows of the embedding matrix.

5.2 Clustering Results in the Original Space

Figure 2 shows the clustering results projected onto the original data space. Both the unnormalized and the normalized spectral clustering approaches successfully identify the three concentric circular clusters, which are non-linearly separable in the original space. By rescaling each node according to its degree, the normalized Laplacian reduces the influence of highly connected nodes and provides a more balanced representation of the data.

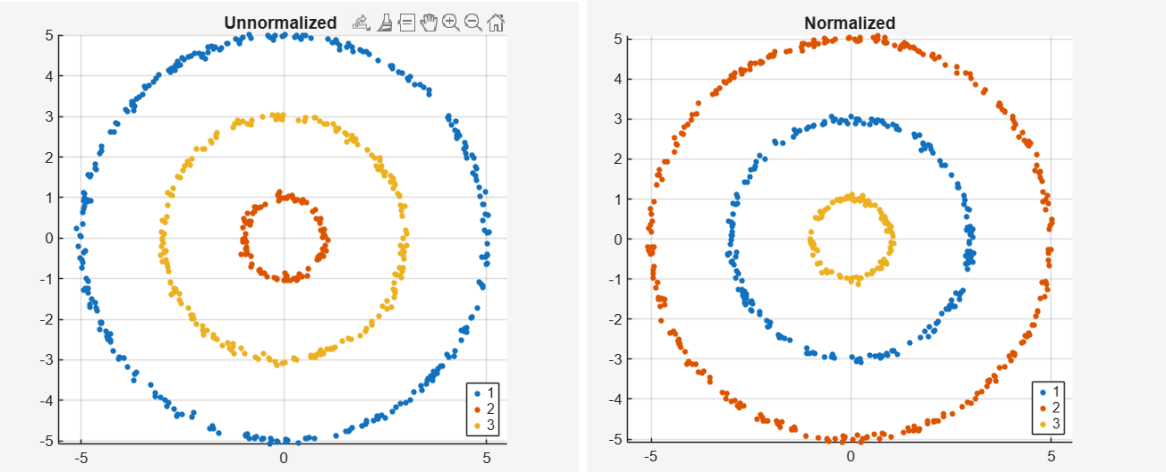


Figure 2: Spectral clustering results on the concentric circles dataset. Unnormalized Laplacian $L = D - W$ (left) and normalized Laplacian L_{sym} (right).

5.3 Spectral Embedding Analysis

Figure 3 shows the spectral embeddings obtained from the two Laplacians. In the unnormalized case, the embedding points tend to collapse toward similar directions, which reflects the sensitivity of the unnormalized Laplacian to degree variations. In contrast, the normalized embedding

exhibits a clearer separation between the clusters, with points distributed more uniformly in the spectral space. This behavior explains why applying k-means in the spectral domain is effective: the spectral embedding transforms the original non-convex structure into a representation where clusters become linearly separable.

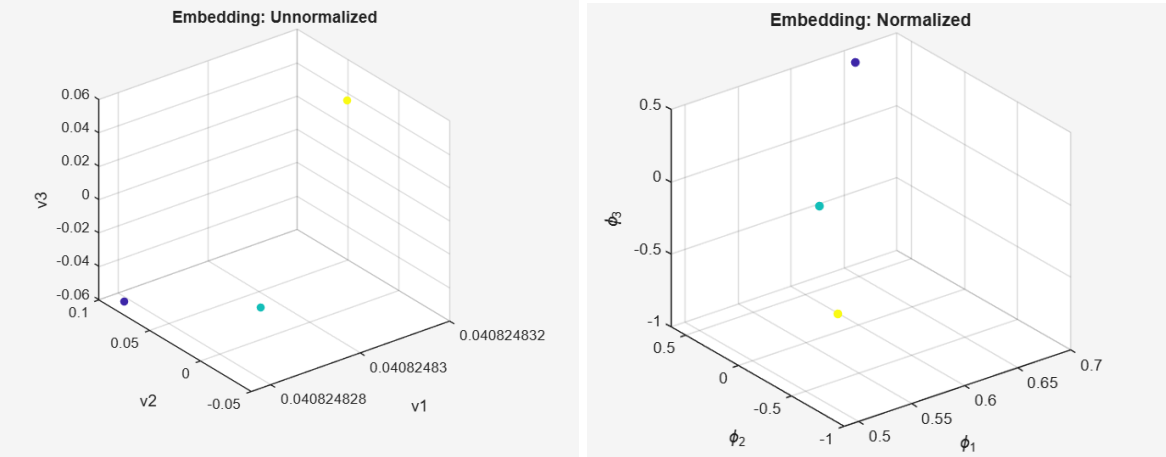


Figure 3: Spectral embeddings obtained using the unnormalized Laplacian (left) and the normalized Laplacian (right).

6 Discussion

The results show that spectral clustering works well on non-convex datasets. In the considered example, the data points are arranged along concentric circles, so standard distance-based clustering methods are not able to correctly separate the clusters in the original space.
Both unnormalized and normalized spectral clustering successfully identify the correct cluster structure. However, the unnormalized Laplacian is more sensitive to variations in node degree, which can lead to less stable results when clusters have different sizes or densities.
The normalized Laplacian instead provides a more balanced spectral embedding, resulting in clearer separation and more robust clustering.
Overall, these results confirm that spectral clustering is effective for complex data structures and that the normalized formulation is generally preferable in practice.